Book Recommender System

for DataSentics

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Chapter 1 – The Past

Matrix Factorization

$$Y_{ai} = \mu + b_a + c_i + \sum_{k=1}^{K} U_{ak} V_{ik}$$

$Y : n_{users} \times n_{books}$

$U : n_{users} \times k$

$V : n_{users} \times k$

$a \ldots user\ index$

$i \ldots book\ index$

$$L = \sum_{(a,i)\in\Omega} \left(Y_{ai} - \mu - b_a - c_i - u_a^T v_i\right)^2 + \lambda_b \left(\sum_a b_a^2 + \sum_i c_i^2\right) + \lambda_f \left(\sum_a \|u_a\|^2 + \sum_i \|v_i\|^2\right)$$

Tuning *k*

k	1	2	4	8	16	32	64	128	256	384
#	38	38	2	5	0	13	5	5	1	1

- number of times (out of 100) when model with interaction terms didn’t outperform model with only biases

Planned Hybrid Loss Function

$$
\begin{aligned}
L = \sum_{(a,i)\in\Omega} \left(Y_{ai} - \mu - b_a - c_i - u_a^T v_i\right)^2 \\
+ \lambda_b \left(\sum_a b_a^2 + \sum_i c_i^2\right) \\
+ \lambda_f \left(\sum_a \|u_a\|^2 + \sum_i \|v_i\|^2\right) \\
+ \lambda_e \left(\sum_i \|v_i - e_i\|^2\right)
\end{aligned}
$$

$e_i \ldots book\ embeddings\ via\ LLM$

Evaluation Metrics

- **Agreement with bias-only:** On average, the share of the same recommended items for a given user as the bias-only model.
- **Mean overlap:** Expected share of shared items between two random users. My main personalization metric.
- **Coverage:** Total number of distinct recommended books.
- **Relative log-ARP:** How popular (in terms of number of ratings) the recommended items are, on average.
- **Relative mean rating:** Average rating of recommended books / μ
- **Entropy:** How evenly the recommendations are distributed across items.

First Models

Model	RMSE	AWB	MOV	COV	ARP	RMR	ENT
random	8.03	0.001	0.001	0.981	-0.66	0.99	0.99
bias-only	1.58	1	0.995	0.001	-0.03	1.23	0.18
full	1.58	0.755	0.801	0.014	0.05	1.23	0.23
full_w_beta	1.77	0.001	0.035	0.158	1.26	0.98	0.64

- full: $$Y_{ai} = \mu + b_a + c_i + \sum_{k=1}^{K} U_{ak} V_{ik}$$

- full_w_beta: $$Y_{ai} = \mu + b_a + \beta c_i + \sum_{k=1}^{K} U_{ak} V_{ik}$$

- tuned for RMSE

Improved Models

Model	NDCG	RMSE	AWB	MOV	COV	ARP	RMR	ENT
random	0.000	8.03	0.001	0	0.98	-0.66	0.99	0.99
bias-only	0.001	1.58	1	0.96	0	-0.03	1.23	0.18
full	0.001	1.58	0.755	0.80	0.01	0.05	1.23	0.23
full_w_beta	0.022	1.77	0.001	0.04	0.16	1.26	0.98	0.64
UV-only	0.040	6.04	0	0.09	0.11	1.82	1.06	0.53

- same models, just tuned for NDCG@5

Chapter 2 – The Present

Final Model(s)

Model	NDCG	AWB	MOV	COV	ARP	RMR	ENT
UV-only.1	0.043	0.001	0.07	0.16	1.68	1.059	0.57
UV-only.2	0.045	0.001	0.05	0.24	1.50	1.059	0.61
UV-only.3	0.046	0.001	0.04	0.34	1.27	1.060	0.67
UV-only.4	0.046	0.002	0.02	0.45	0.99	1.060	0.72
UV-only.5	0.044	0.003	0.02	0.56	0.74	1.060	0.77

$$Y_{ai} = \sum_{k=1}^{K} U_{ak} V_{ik} - \gamma \log(popularity)$$

Harry Potter Smoke Test

Inputs:

- 0590353403,Harry Potter and the Sorcerer's Stone (Book 1),J. K. Rowling,1998,Scholastic
- 0439064864,Harry Potter and the Chamber of Secrets (Book 2),J. K. Rowling,1999,Scholastic
- 0439136350,Harry Potter and the Prisoner of Azkaban (Book 3),J. K. Rowling,1999,Scholastic
- 0439139597,Harry Potter and the Goblet of Fire (Book 4),J. K. Rowling,2000,Scholastic

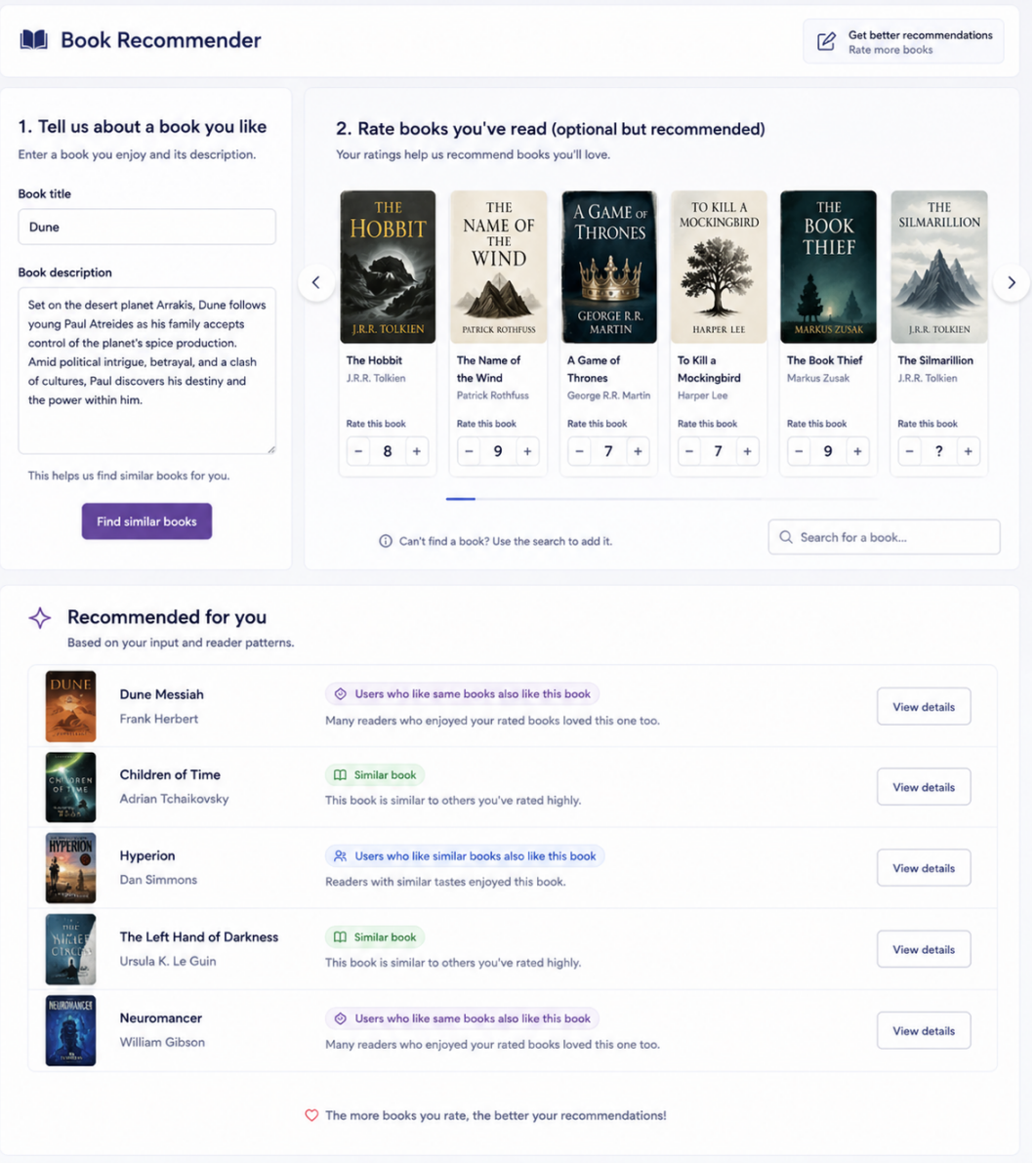
Results:

- **043935806X, Harry Potter and the Order of the Phoenix (Book 5),J. K. Rowling,2003,Scholastic**
- 059035342X, Harry Potter and the Sorcerer's Stone (Harry Potter (Paperback)),J. K. Rowling,1999,Arthur A. Levine Books
- 0439139600, Harry Potter and the Goblet of Fire (Book 4),J. K. Rowling,2002,Scholastic Paperbacks
- 0439064872, Harry Potter and the Chamber of Secrets (Book 2),J. K. Rowling,2000,Scholastic
- 0439136369, Harry Potter and the Prisoner of Azkaban (Book 3),J. K. Rowling,2001,Scholastic

Chapter 3 – The Future

- Embeddings for content-based filtering
- Using implicit ratings only (SDG)
- Solving the cold-start problem

Vision of the UI



Thank you for your attention